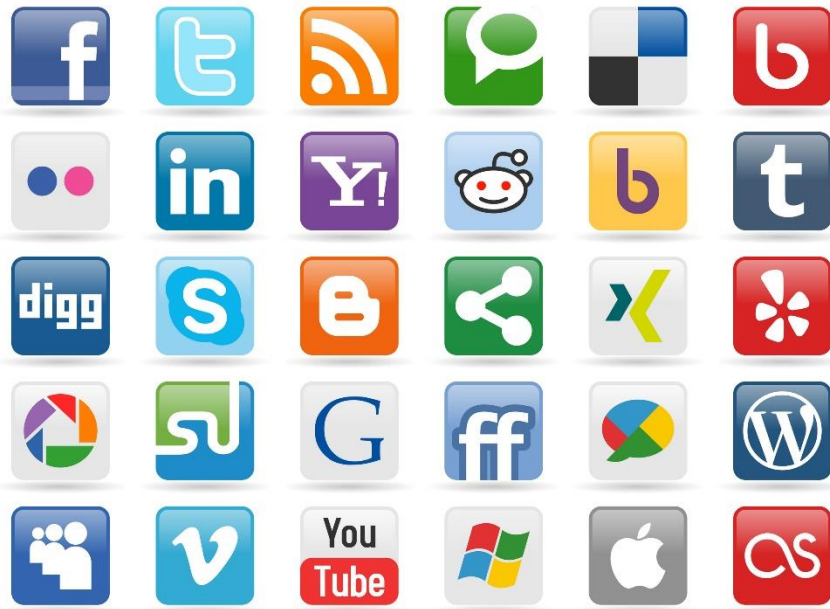
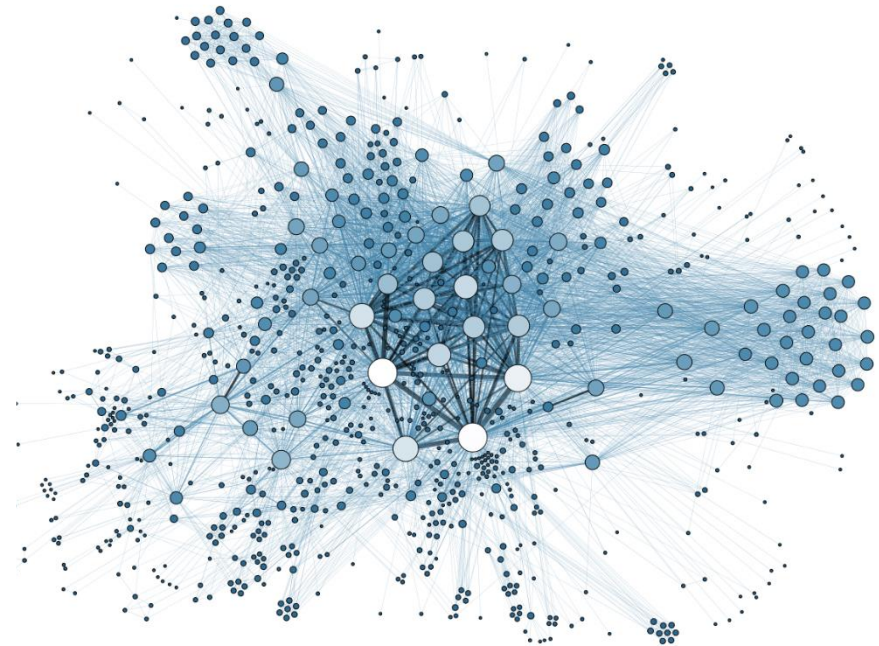




SE4095-Social Network: Theory and Practice

Introduction



Course Information

- SE4095-Social Network: Theory and Practice
- Text Book: Zafarani, R., Abbasi, M. A., & Liu, H. (2014). Social media mining: an introduction. Cambridge University Press.

Course Information

DESCRIPTION

Social network analysis is focused on uncovering and understanding the hidden patterns in social networks. Network analysis is based on the intuitive notion that the patterns of social structure are important features of the lives of the individuals who display them. This course focuses on basic concepts in social networks. This course also focuses on the recent research trends in social networks research. The use of modern social network analysis techniques will also be covered.

Course Information

DESCRIPTION & OBJECTIVES

After the successful completion of the course, the students will be able to:

- Understand the social aspects of Social Networks
 - Social Theories + Social Media + Mining
 - Learn to collect, clean, and represent social network data
 - How to measure important properties of social networks
 - Find and analyze communities in social networks
 - Understand how information propagates in social networks
 - Understanding friendships, performing recommendations, and analyzing behavior
- Study or ask about interesting projects/ research issues
 - e.g., start-up ideas/research challenges
- Use representative algorithms and tools

The weekly schedule
will be available on
Google Classroom

7A- 42i6zugu

7B- bkvoackd

Course Workload and Evaluation

- **A lot** of work is expected from you!
 - Lectures
 - Slides
 - Additional Material such as research articles
 - Homework assignments
 - Conceptual; deep thinking required
 - Coding based
 - Projects
 - 1 projects
 - Group based (2 members)
 - 3 Exams – Mid 1, Mid 2, Final
 - Quizzes –after chapters/topics
 - Class Activity – can be in each class
 - Assignment Late penalty:
 - Exponential penalty: -50% first day, -75% second day, and no points on or after the third day.

Social Media

Definition

Social Media is the use of electronic and Internet tools for the purpose of sharing and discussing information and experiences with other human beings in more efficient ways.

Social Media Landscape 2015



Facebook

The image shows a screenshot of Mark Zuckerberg's Facebook profile. The page layout includes a top navigation bar with the Facebook logo, a search bar, and links to Home, Profile, and Account. On the left is a sidebar with a large profile picture of Mark Zuckerberg and links to Wall, Info, Photos (826), Questions, and Family. The main content area displays his name, a bio mentioning his work at Facebook, education at Harvard University, and residence in Palo Alto, California. Below the bio are several photo albums. The 'Education and Work' section lists his employers (Facebook), college (Harvard University), and high school (Ardsley High School). The 'Family' section lists his mother, father, and three sisters. The 'Philosophy' section features a quote about children and artists. On the right side, there are links to 'Send Message' and 'Poke', a section for 'You and Mark' with mutual friends, and a 'Sponsored' section with various advertisements.

- How does Facebook use your data?
- Where do you think Facebook can use your data?

What about Amazon?

The screenshot shows the Amazon.com homepage with a browser window at the top. The address bar displays "www.amazon.com". The main navigation bar includes the Amazon logo, links for "Your Amazon.com", "Today's Deals", "Gift Cards", and "Help". On the right, it promotes "FREE Two-Day Shipping" with a link to "Join Amazon Prime Today". Below this, there are links for "Shop by Department", a search bar with a dropdown menu set to "All", a "Go" button, and links for "Hello. Sign in Your Account", a shopping cart icon with "0" items, and a "Wish List".

A horizontal banner for "Amazon Instant Video: The latest TV shows, available instantly." is followed by a link to "EXPAND TO LEARN MORE". Below this, a row of category links includes "Instant Video", "MP3 Store", "Cloud Player", "Kindle" (highlighted), "Cloud Drive", "Appstore for Android", "Digital Games & Software", and "Audible Audiobooks".

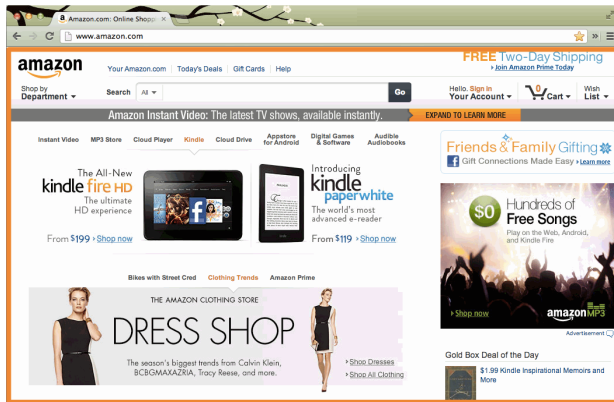
The main content area features two product highlights for Kindle devices. On the left, the "The All-New kindle fire HD" is advertised as "The ultimate HD experience", priced "From \$199" with a "Shop now" link. It shows a tablet displaying a Facebook interface. On the right, the "Introducing kindle paperwhite" is advertised as "The world's most advanced e-reader", priced "From \$119" with a "Shop now" link. It shows a tablet displaying a book page.

To the right of these highlights is a "Friends & Family Gifting" section with a Facebook logo and the text "Gift Connections Made Easy" and a "Learn more" link. Below this is a large advertisement for "Hundreds of Free Songs" with a "\$0" price tag, stating "Play on the Web, Android, and Kindle Fire". It features an image of a crowd at a concert and a "Shop now" link. The Amazon MP3 logo is also present.

At the bottom left, there is a section for "THE AMAZON CLOTHING STORE" titled "DRESS SHOP". It features two models in black dresses. Text below the models reads: "The season's biggest trends from Calvin Klein, BCBGMAXAZRIA, Tracy Reese, and more." There are two links: "Shop Dresses" and "Shop All Clothing".

At the bottom right, there is a "Gold Box Deal of the Day" for "\$1.99 Kindle Inspirational Memoirs and More", accompanied by an image of a book cover.

What about Amazon?



Although Amazon is not a traditional social networking platform like Facebook, it still uses network analysis in several ways:

- **Recommendation Networks:** Connect users with products based on similar purchasing behavior (collaborative filtering).
- **Product Networks:** Identify products that are frequently bought together.
- **Customer Similarity Networks:** Group customers with similar interests and purchasing patterns.
- **Review Networks:** Analyze reviewer interactions and product ratings.
- **Seller-Buyer Networks:** Study relationships between customers, sellers, and products.

Example:

- If Customer A and Customer B purchase many of the same products, Amazon considers them similar.
- If Customer A buys a new product that Customer B has not seen, Amazon may recommend it to Customer B.

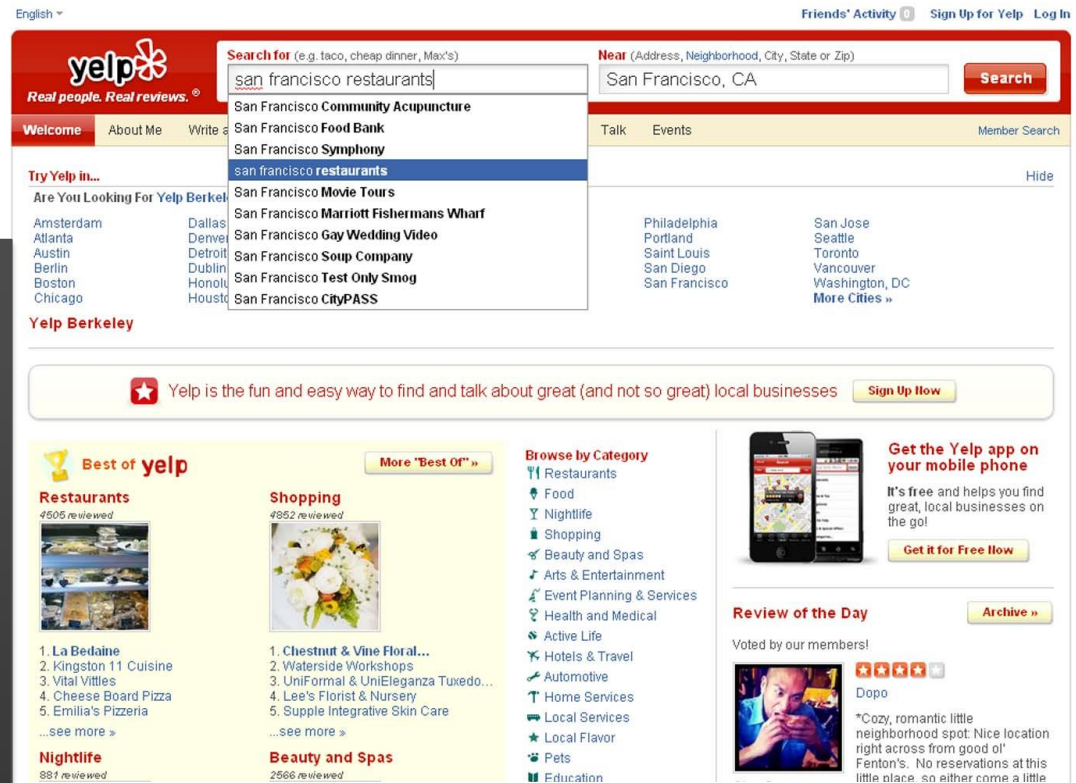
This forms a **customer-product network**

Yelp?

Yelp is a location-based review and recommendation platform where users discover, review, and rate local businesses such as restaurants, hotels, cafes, salons, and repair services.

What data does Yelp collect?

- Name and profile information
- Location (GPS or city)
- Search history
- Reviews and ratings
- Photos uploaded
- Check-ins
- Business bookmarks or favourites
- Device and browser information
- Clicks and browsing behaviour



Yelp?



Yelp is an excellent example of a **social network application** because it contains interactions among **users**, **businesses**, and **reviews**.

The relationships can be modeled as a network:

- **User** → **Business**: writes a review or gives a rating.
- **User** → **User**: follows friends, trusts reviewers, or likes helpful reviews.
- **Business** → **Category**: belongs to categories such as restaurants, hotels, or shopping.
- **User** → **Location**: searches and checks in at different places.

Using Social Network Analysis, Yelp can:

- Identify **influential reviewers** using centrality measures.
- Detect **communities** of users with similar tastes.
- Recommend restaurants and local businesses based on similar users' preferences.
- Detect **fake review networks** and suspicious reviewer behavior.
- Analyze trends in customer opinions across different locations.

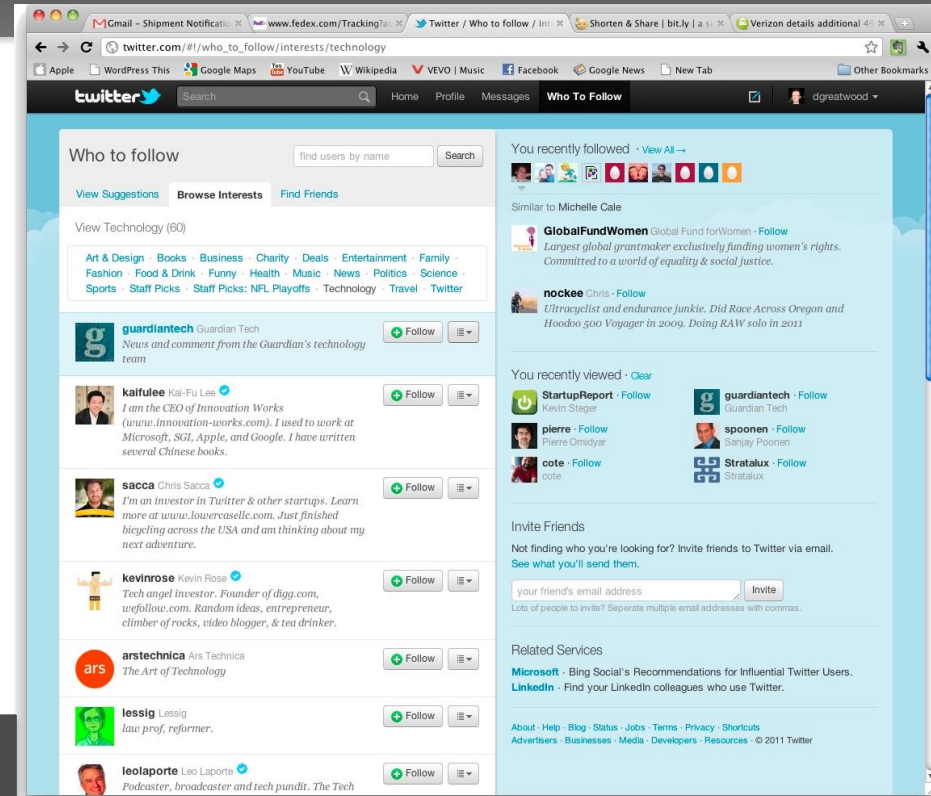
Example

- Suppose you frequently review **Italian restaurants** and give high ratings to **pizza places**. Yelp compares your preferences with users who have similar review patterns. If those users highly recommend a new Italian restaurant that you haven't visited, Yelp may recommend it to you.

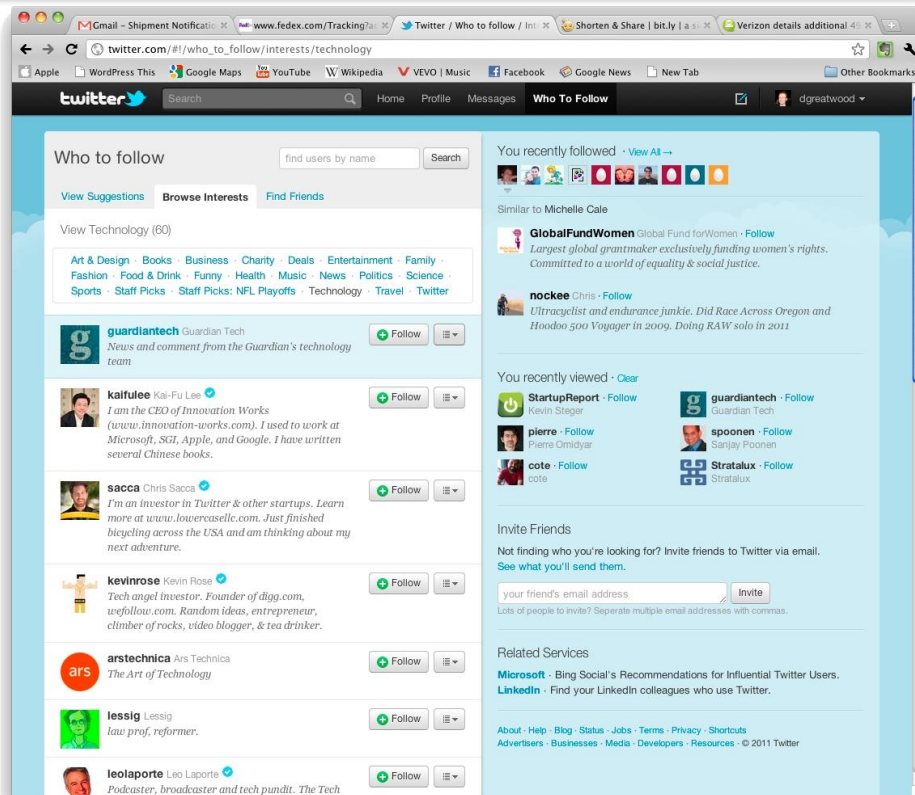
Or Twitter?

What data does Twitter collect?

- Profile information
- Followers and following lists
- Tweets, retweets, replies, and likes
- Hashtags and mentions
- Search history
- Topics you follow
- Location (if enabled)
- Device and browser information
- Time spent viewing tweets



Or Twitter?



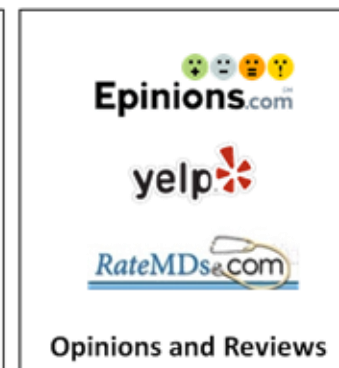
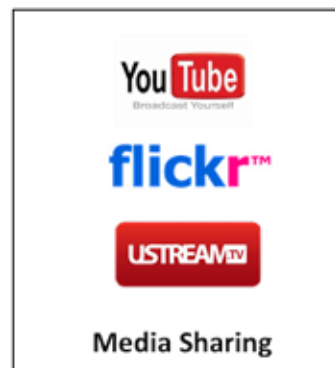
- Social Network Analysis is used to:
- Measure **user influence** using centrality measures (e.g., PageRank, betweenness, eigenvector centrality).
- Detect **communities** based on shared interests and interactions.
- Recommend **people to follow** using mutual connections and similarity.
- Analyze **information diffusion**, such as how tweets, hashtags, or news spread through the network.
- Identify **opinion leaders** and influential users.
- Detect **fake accounts**, spam networks, and coordinated campaigns.

Social Media: Examples

- A wiki article
- Web reviews and ratings of a popular pizza place in your city
 - E.g., Yelp.com
- An online social network of your professional contacts
 - E.g., Facebook.com, LinkedIn.com
- An iPhone application that informs you where parking is likely available
 - FasPark

Types of Social Media

- Online Social Networking
- Publishing
 - Blogging
 - Wiki
- Micro blogging
- Social News
- Social Bookmarking
- Media Sharing
 - Video Sharing
 - Photo Sharing
 - Podcast Sharing
- Opinion, Review, and Ratings Websites
- Answers
- Entertainment



Online Social Networking

Category	Description	Popular Examples
Online Social Networking	Platforms for connecting with people and building social relationships.	Facebook, LinkedIn, Instagram, Threads
Publishing	Platforms for publishing articles, documents, or written content.	Medium, WordPress, Google Sites
Blogging	Websites where individuals or organizations regularly post articles or blogs.	WordPress, Blogger, Medium, Tumblr
Wiki	Collaborative websites where users create and edit content collectively.	Wikipedia, Wikibooks
Microblogging	Platforms for sharing short messages, updates, or posts.	X (Twitter), Threads
Social News	Users submit, vote on, and discuss news stories.	Reddit
Social Bookmarking	Save, organize, and share web pages or resources.	Pinterest, Flipboard
Media Sharing	Share different types of digital media (photos, videos, audio, documents).	Instagram, Flickr
Video Sharing	Upload, watch, and share videos.	YouTube, TikTok, Dailymotion
Photo Sharing	Share and organize photographs.	Instagram, Flickr, Google Photos
Podcast Sharing	Publish, distribute, and listen to podcasts.	Spotify, Apple Podcasts, Google Podcasts(legacy)
Opinion, Review, and Ratings Websites	Users review products, restaurants, hotels, movies, etc.	Yelp, Tripadvisor, IMDb, Google Reviews
Answers (Q&A)	Ask questions and receive answers from the community or experts.	Quora, Stack Overflow, Stack Exchange, Yahoo! Answers (discontinued)
Entertainment	Platforms for streaming, gaming, or entertainment content.	Netflix

Main Characteristics

- **Participation**
 - social media encourages contributions and feedback from everyone who is interested. It blurs the line between media and audience.
- **Openness**
 - most social media services are open to feedback and participation. They encourage voting, comments and the sharing of information. There are rarely any barriers to accessing and making use of content – password-protected content is frowned on.
- **Conversation**
 - whereas traditional media is about “broadcast” (content transmitted or distributed to an audience) social media is better seen as a two-way conversation.
- **Community**
 - social media allows communities to form quickly and communicate effectively. Communities share common interests, such as a love of photography, a political issue or a favorite TV show.
- **Connectedness**
 - Most kinds of social media thrive on their connectedness, making use of links to other sites, resources and people.

Main Characteristics

Characteristic	Simple Meaning	Example
Participation	Everyone can create, like, comment, and share content.	Facebook likes, YouTube comments
Openness	Information is openly shared and users can provide feedback.	Wikipedia editing, Amazon reviews
Conversation	Users communicate with each other instead of only receiving information.	Twitter/X replies, Facebook comments
Community	People with common interests form groups and interact.	Reddit, Facebook Groups, Discord
Connectedness	People, content, websites, and platforms are linked together.	Instagram linking to YouTube, Facebook, or an online store

Social Network Analysis, or SNA, *is a method used to visualize and analyse relationships and connections between entities or individuals within a network.*

SNA studies relationships, not just individuals.

- Imagine mapping the relationships between different departments in a corporation.
- The outcome would be a vivid picture of how each department interacts with others, allowing us to see communication patterns, influential entities, and bottlenecks

Importance of SNA

- SNA is a powerful method.
- It allows us to explore the underlying structure of an organization or network, identifying the formal and informal relationships that drive the formal processes and outcomes.
- This insight can enable better communication, facilitate change management, and inspire more efficient collaboration.
- This methodology also helps demonstrate the impact of relationship-building.
- The maps and visualizations produced by SNA are an engaging way to share your progress and impact with stakeholders, donors, and the community at large.

Importance of SNA

1. SNA is a powerful method.

- Instead of looking at people individually, it studies **how everyone is connected**.
- Example:
- Facebook doesn't just know you.
- It knows: Your friends, Their friends, Groups, Pages, Likes, Comments
- This network provides valuable insights.

2. It reveals hidden relationships.

- Organizations have:
- Formal relationships (Manager → Employee)
- Informal relationships (Employees from different departments helping each other.)
- Sometimes informal communication is stronger than formal communication.
- SNA discovers these hidden connections.

3. Improves Communication

- Suppose HR wants to share an announcement.
- SNA can identify the employees who communicate with the largest number of colleagues.
- Instead of emailing everyone,
- HR can first inform these influential employees.
- The information spreads much faster.

Importance of SNA

4. Helps Change Management

- Imagine a university introducing a new LMS.
- Some faculty members are highly respected.
- If they support the new LMS,
- others are more likely to adopt it.
- SNA identifies these influential people.

5. Encourages Better Collaboration

- Suppose two research groups have never collaborated.
- SNA may reveal that both frequently interact with the same professor.
- That professor can connect the groups, leading to collaboration.

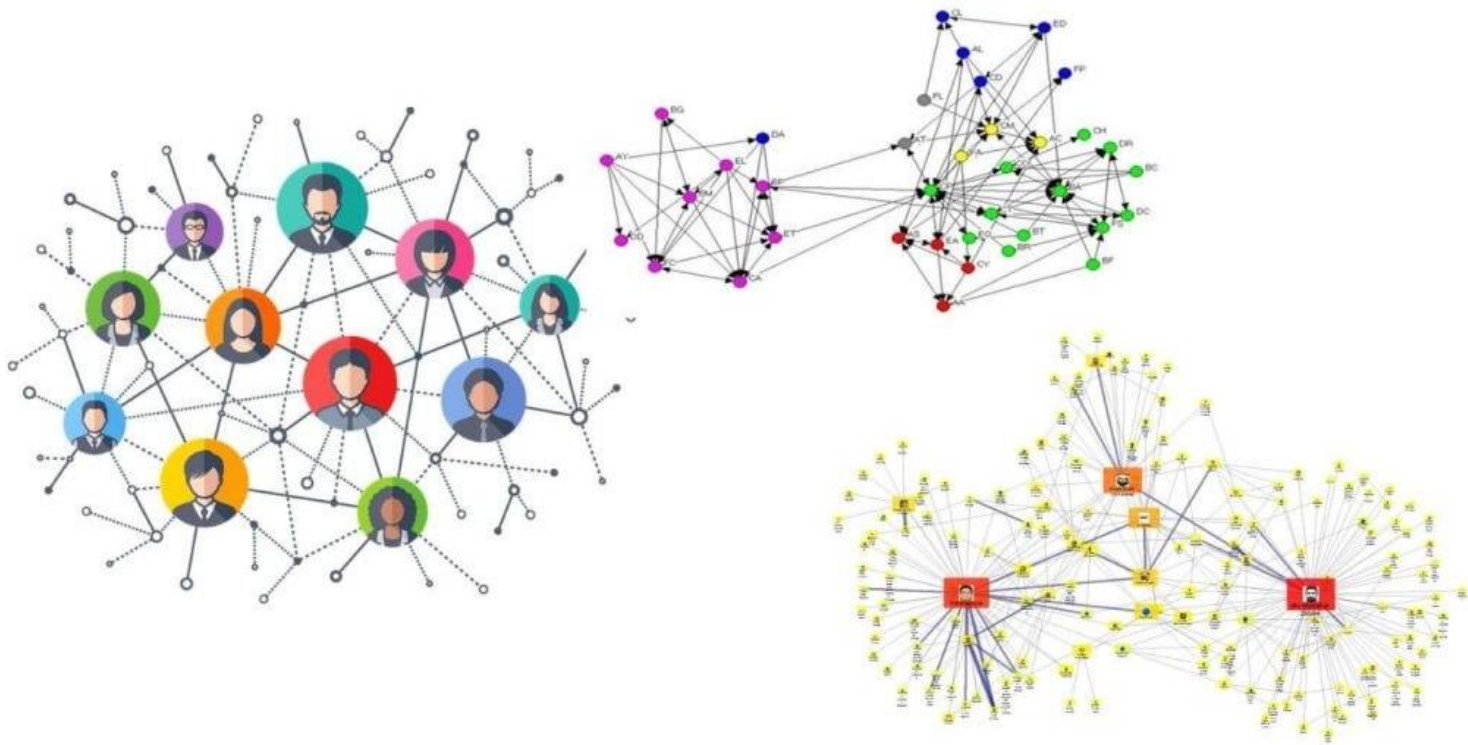
6. Easy-to-understand Visualization

- Instead of reading hundreds of rows in a spreadsheet,
- SNA draws a graph.

Fundamentals of SNA

- **Nodes and Edges**

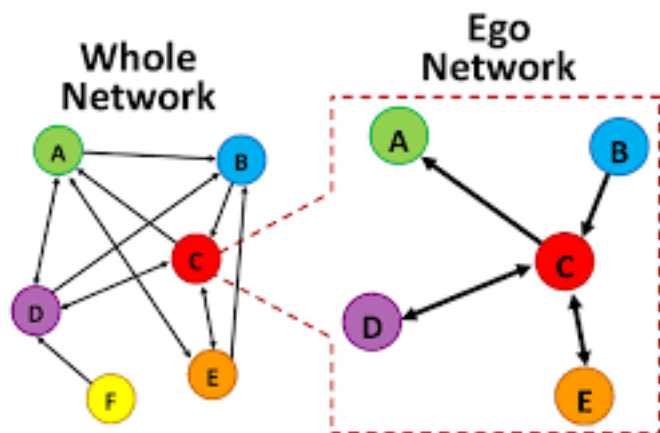
- In SNA, nodes represent individuals or entities while edges symbolize the relationships between them. For example, in an inter-organizational network, nodes might be companies, and edges could represent communication, collaboration, or competition.



Fundamentals of SNA

- **Network Types**

- Different types of networks serve different purposes.
 - 'Ego Networks' focus on one node and its direct connections, revealing its immediate network.
 - 'Whole Networks', on the other hand, capture a broader picture, encompassing an entire organization or system.
 - Open networks are loosely connected, with many opportunities to build new connections, ideal for innovation and idea generation – while closed networks are densely interconnected, and better for refining ideas amongst a group who all know each other.



Fundamentals of SNA

- **Network Properties**

- Properties such as

- **density** (the proportion of potential connections that are actual connections),
 - **diameter** (the longest distance between two nodes)
 - **centrality** (the importance of a node within the network)

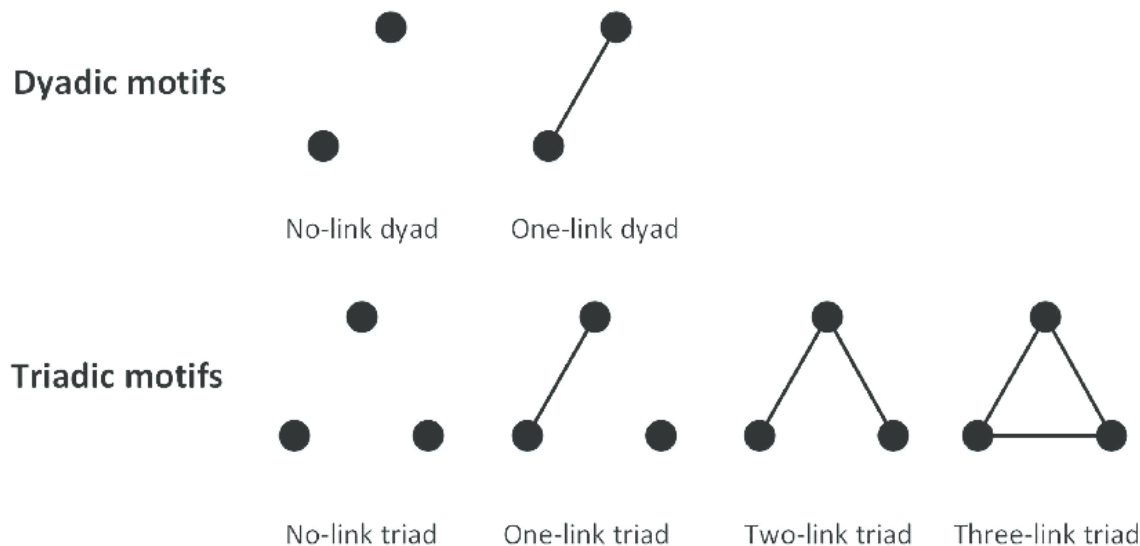
- allow us to understand the network's structure and function.

- Metrics also can measure relationship quality across the network, like our validated trust and value scores.

Fundamentals of SNA

- **Dyadic and Triadic Relationships**

- Dyadic relationships involve two nodes, like a partnership between two companies.
- Triadic relationships, involving three nodes, are more complex but can offer richer insights.
- For instance, it might show how a third company influences the relationship between two others, or which members of your network are the best at building new relationships between their peers.

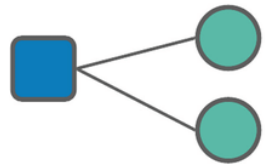


Fundamentals of SNA

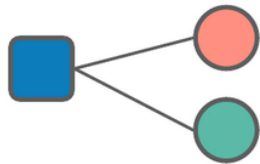
- **Homophily and Heterophily**

- Homophily refers to the tendency of similar nodes to connect, while heterophily is the opposite or diverse.
 - In a business context, we might see homophily between companies in the same industry
 - heterophily when seeking diversity in users.
- Many networks aim to be diverse but get stuck talking to the same, similar partners. These network concepts underly many strategies promoting network innovation to avoid groupthink among likeminded partners.

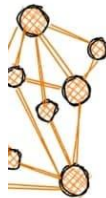
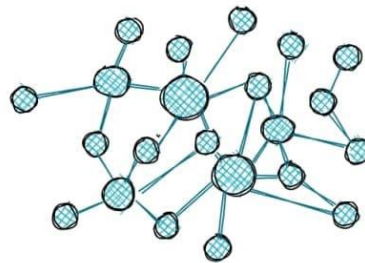
Homophily



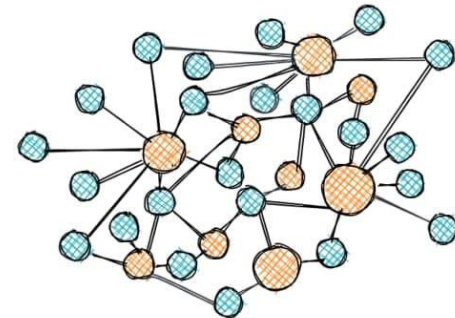
Heterophily



Homophily

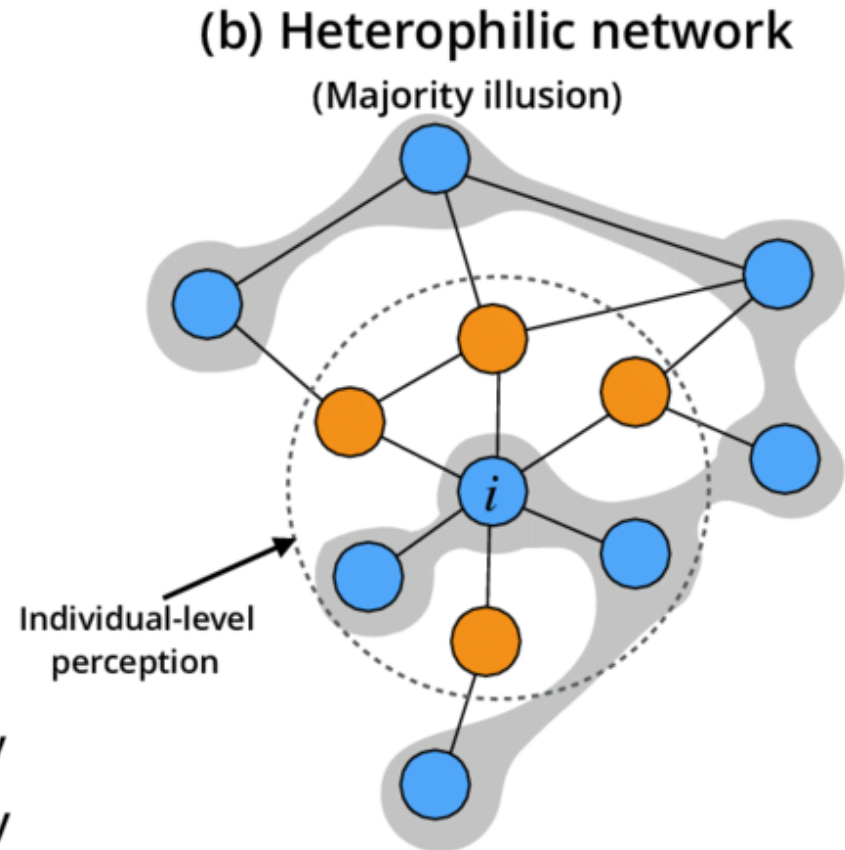
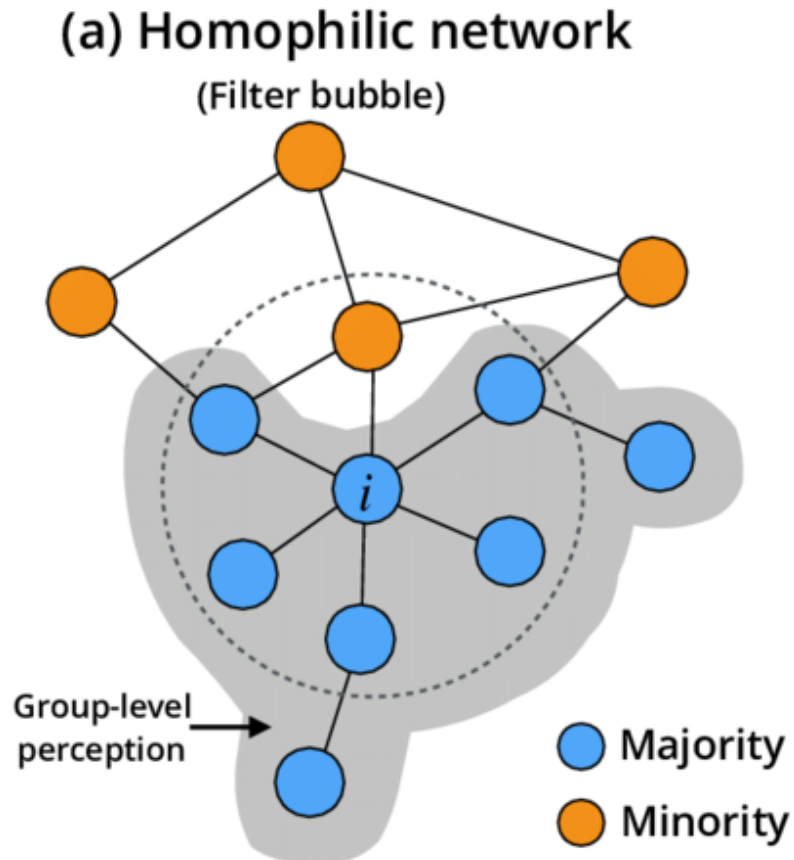


Heterophily



Fundamentals of SNA

- **Homophily and Heterophily**
 - Group Level



Fundamentals of SNA

- **Network Topologies**

- Lastly, the layout or pattern of a network, its topology, can reveal much about its function.
- For instance, a centralized topology, where one node is connected to all others, may indicate a hierarchical organization, while a decentralized topology suggests a more collaborative and flexible environment.

Social Media Mining is the process of representing, analyzing, and extracting meaningful patterns from social media data

Social Media Mining Challenges

1. **Big Data Paradox**

1. Social media data is big, yet not evenly distributed.
2. Often little data is available for an individual

2. **Obtaining Sufficient Samples**

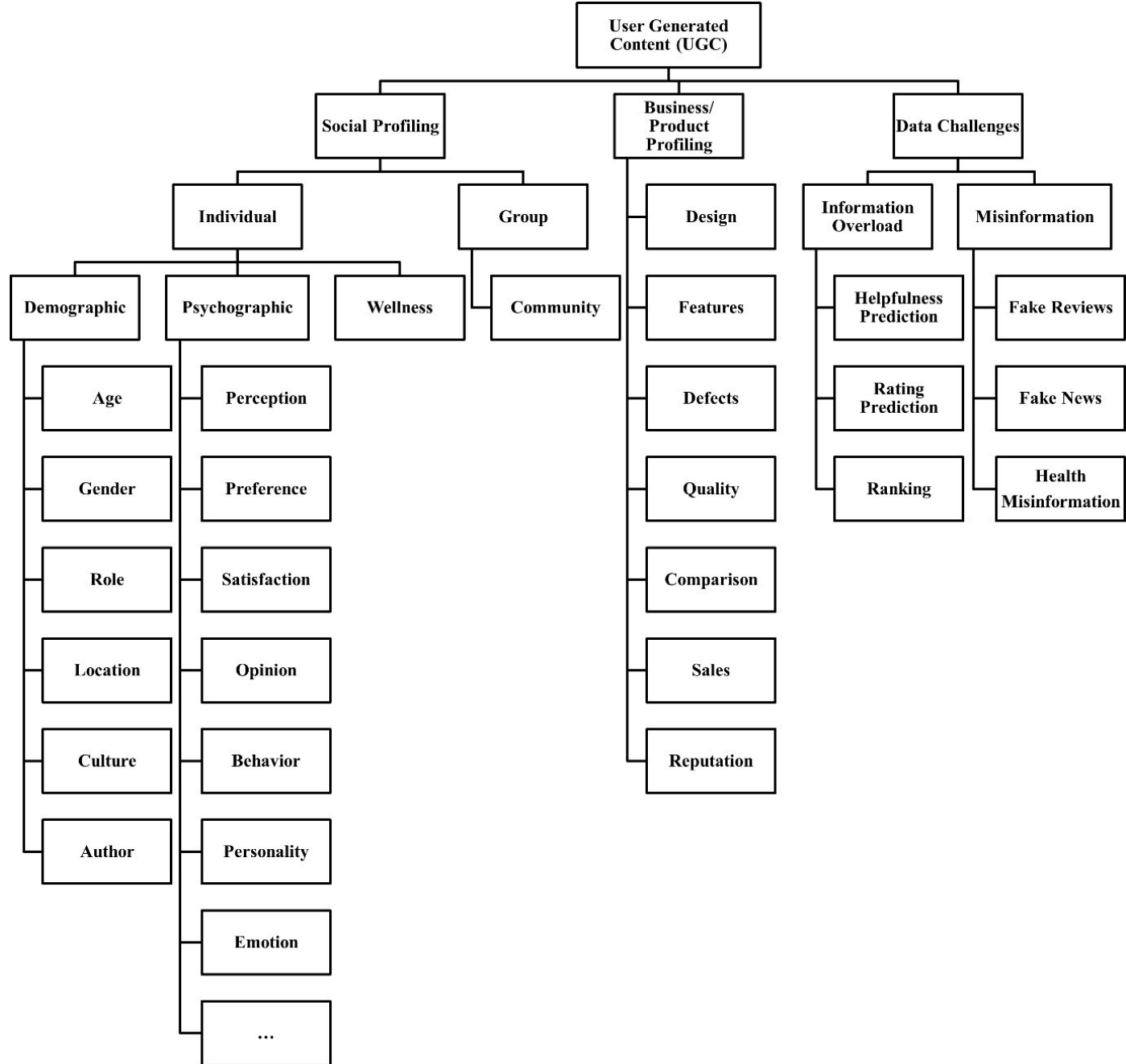
1. Are our samples reliable representatives of the full data?

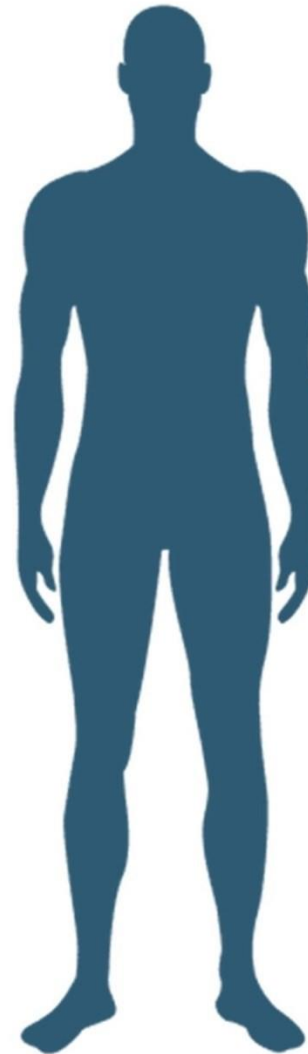
3. **Noise Removal Fallacy**

1. Too much removal makes data more sparse
2. Noise definition is relative and complicated and is task-dependent

4. **Evaluation Dilemma**

1. When there is no ground truth, how can you evaluate?



GROUP**INDIVIDUAL****DEMOGRAPHICS**

Age, Gender
Education
Income, Occupation
Marital Status
Religion
Race / Ethnicity
Language
Region/Location
City, Nationality

PSYCHOGRAPHICS

Personality
Behavior, Emotions
Mood, Habits
Social Influence
Preference
Social Connections
Activity, Interests
Opinions, Attitudes
Values

WELLNESS

Body Mass Index
Disease Tendency

Recent Advances & Research Trends, Application Domains, Open research issues (1)

- Information Overload
- Review Helpfulness
- Rating
- Ranking
- Recommendation
- Mis/Disinformation
- Fake News
- Fake Reviews
- Health Misinformation
- Social Profiling

Recent Advances & Research Trends, Application Domains, Open research issues (2/3)

- Election Prediction
- Behavior
- Opinion Mining
- Cyberbullying
- Community Detection
- Trajectories Detection
- Knowledge Discovery/Topic Modeling
- Detection of Viral Posts
- Homophily social networks
- Information diffusion
- Influence maximization

Recent Advances & Research Trends, Application Domains, Open research issues (3/3)

- Community detection
- Sentiment Analysis
- Pattern Mining
- ...

- **Information Overload:** SNA helps filter and prioritize overwhelming data by identifying key nodes and information diffusion patterns.
- **Review Helpfulness:** SNA assesses review credibility and influence by analyzing reviewer connections and the network's feedback loops.
- **Rating:** SNA leverages user interconnections to predict and improve rating accuracy through trust-based and community-driven approaches.
- **Ranking:** SNA enhances content ranking by evaluating node influence, centrality, and the spread of information within a network.
- **Recommendation:** SNA drives personalized recommendations by analyzing user similarity, social ties, and information flow in the network.
- **Mis/Disinformation:** SNA identifies and mitigates the spread of false information by tracking its propagation through the network.
- **Fake News:** SNA detects and combats fake news by mapping the network of sources, sharers, and influencers spreading it.
- **Fake Reviews:** SNA uncovers fake reviews by analyzing reviewer behavior, network clustering, and suspicious activity patterns.
- **Health Misinformation:** SNA helps combat health misinformation by tracing its spread and identifying influential nodes for targeted interventions.
- **Social Profiling:** SNA constructs detailed user profiles by examining social connections, interactions, and the structure of social networks.

- **Election Prediction:** SNA predicts electoral outcomes by analyzing voter interactions, sentiment, and opinion leaders within social networks.
- **Behavior:** SNA examines user behavior patterns by exploring network structures and the influence of peers on individual actions.
- **Opinion Mining:** SNA extracts and analyzes opinions from social media by evaluating the sentiments expressed within user networks.
- **Cyberbullying:** SNA detects and mitigates cyberbullying by identifying aggressive behavior patterns and influential nodes in the network.
- **Community Detection:** SNA identifies clusters or communities within networks, revealing subgroups with shared interests or connections.
- **Trajectories Detection:** SNA tracks and predicts movement patterns by analyzing how entities interact and move within a network over time.
- **Knowledge Discovery/Topic Modeling:** SNA uncovers hidden topics and patterns by mapping and analyzing the flow of information across the network.
- **Detection of Viral Posts:** SNA identifies and predicts which posts will go viral by analyzing the spread and engagement within the network.

- **Homophily in Social Networks:** SNA explores the tendency of similar individuals to connect and form clusters within social networks.
- **Information Diffusion:** SNA studies how information spreads through networks, identifying key influencers and pathways.
- **Influence Maximization:** SNA optimizes the spread of influence by identifying the most influential nodes and strategic points within the network
- **Community Detection:** SNA identifies and maps out groups or clusters within networks, revealing substructures and shared interests among users.
- **Sentiment Analysis:** SNA analyzes emotional tones and opinions within social networks by evaluating sentiment propagation and influential nodes.
- **Pattern Mining:** SNA discovers recurring behavioral or interaction patterns within social networks, aiding in understanding complex network dynamics.

TODO Items

- 4 To-do items for you:
 - Chapter 1 (read)
 - Install Jupiter Notebook/Anaconda, Gephi, and NetworkX
 - Make a group of your Project (2 members)
 - Think of a topic for your project to work on (considering the research issues discussed or any other)

15-Minute Group Activity: Social Network Analysis

Group size: 3–4 students

Task

- Create a social network of **12 fictional students**. Your network must follow these rules:
- Each student must have **at least 2 connections**.
- Each student can have **maximum 5 connections**.
- Create **at least 2 distinct communities/groups**.
- Include **at least one student who connects two communities**.
- Draw the network using **nodes and edges**.

Questions

1. **Who is the most powerful person in your network?** What makes them powerful?
2. **Who is the “connector” of your network?** Which person connects two otherwise separate groups?
3. **Who would you trust to spread an important message to everyone?** Why?
4. **Which person would cause the biggest disruption if they suddenly disappeared from the network?**
5. **Can you identify at least two different communities in your network?** What makes them communities?
6. **Who is the most isolated person?** How could you help connect them to the network?
7. **Is your network more “birds of a feather” (homophily) or “opposites attract” (heterophily)?** Give an example.
8. **If you could add only ONE new friendship, which two people would you connect?** Why would this improve the network?
9. **If you could remove ONE connection without removing a person, which connection would you remove?** What would happen?
10. **Imagine a rumor starts with one person. Who would you choose as the starting point to make it spread fastest?**
11. **Imagine a new student joins the class. Who should they connect with first to become part of the network quickly?**
12. **If your network were Facebook/Instagram, who would be the “influencer” and who would be the “hidden influencer”? Why?**